Higher National Diploma in Engineering (Civil/QS)

1st Year, Second Semester Examination -2019

QS12022/CE1206 Construction Materials

Instructions for Candidate:

Answer 04 questions only

No. of Questions: 05
No. of Pages : 02
Time: 02(hours)

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1. Concrete is a man-made composite material, which is frequently used in the construction field.

   i. What are the ingredients in each of the following?

      Grout -

      Mortar -

      Concrete -

      Reinforced concrete - (1 x 04 Marks)

   ii. Name 4 types of chemical admixtures used in concrete and briefly explain the performance of each of them. (2 x 04 Marks)

   iii. What is 'workability' of concrete? Name two workability tests. (03 Marks)

   iv. What is meant by the 'grade of concrete'? (02 Marks)

   v. Explain why do we use reinforced steel inside concrete. (02 Marks)

   vi. What is the importance of curing of concrete? (02 Marks)

   vii. Name 4 concrete curing methods followed at the site? (02 Marks)

   viii. What is 'shotcrete'? (02 Marks)

2. Cement and timber are two commonly used building materials.

   i. Name 4 oxides present in cement. (04 Marks)

   ii. Name 3 types of cement, according to ASTM classification and their applications. (06 Marks)

   iii. What is the apparatus used to find setting time of cement? (01 Marks)

   iv. What is meant by 'soundness' of cement? (03 Marks)

   v. What is called 'seasoning' of timber? (03 Marks)

   vi. Explain briefly 'exogenous' and 'endogenous' trees? (04 Marks)

   vii. Name 2 advantages and 2 disadvantages of using timber over concrete, as structural members in building construction. (04 Marks)

3. Construction aggregate, or simply "aggregate", is a broad category of coarse to medium grained particulate material used in construction.

i. Name 3 types of coarse aggregates used in concrete, based on mineral classification. (03 Marks)

ii. Name 2 alternative fine aggregates that can be used for conventional sand. (02 Marks)

iii. Briefly explain what is 'light-weight aggregates'? (03 Marks)

iv. Name 3 light-weight aggregates used in the industry. (03 Marks)

v. Explain briefly what is 'SSD aggregates'? (03 Marks)

vi. What is defined as 'absorption' of aggregates? (03 Marks)

vii. A sand sample weighed 310g when wet and 300g after being dried by an oven. Its weight in SSD condition was measured to be 304g. Determine the absorption (%) and surface moisture content (%) of the sand sample. (08 Marks)

4.

i. Name 2 types of lime available in the market, that is used for construction activities. (04 Marks)

ii. What is 'Slaking of lime'? (02 Marks)

iii. Name 4 usages of lime. (04 Marks)

iv. Name 4 tests for lime (04 Marks)

v. Briefly explain what are 'cast iron' and 'wrought iron'? (04 Marks)

vi. Name 4 factors affecting steel corrosion in RCC? (04 Marks)

vii. Name 3 protection methods against steel corrosion in RCC?

viii. (03 Marks)

5. Bricks and blocks are components of durable masonry construction in which uniformly shaped individual units are laid in courses with mortar as the bed and binding material.

i. Briefly explain the manufacturing process of clay bricks. (04 Marks)

ii. Name 4 tests on bricks. (04 Marks)

iii. Explain how to perform one of the above tests at the site. (03 Marks)

iv. Name 4 advantages of using bricks over other construction materials. (04 Marks)

v. Explain the term 'tempering' in clay brick preparation. (02 Marks)

vi. Name 4 main bond patterns in clay brick masonry, with sketches. (08 Marks)